

Randomization Tests

STAT 145

RANDOMIZATION TESTS

Today's goals

Activity

Consider the following scenario: A sample of 80 four-to six-year olds participated in a study in which they were asked to taste two cereals, both called “Oat Bits” (in random order). The cereal in each box was exactly the same, but the boxes differed in the fact that one of the boxes had pictures of a star mascot and the other did not.



Figure 1: The two cereal boxes shown to the children. The cereal inside was identical; only one box featured a cartoon star mascot

The hypothesis testing framework

Hypothesis testing

Null and alternative hypotheses

The shape of a hypothesis

The question is about...	Null hypothesis ("no effect")	Alternative (pick one direction)	A question of this kind
one proportion			Do more than half of children pick the cartoon box? ()
one mean			Do the bags really average the 16 oz on the label? ()
a difference in proportions			Do two groups differ in who skips the video?
a difference in means			Is average income higher in

Class Example 7.1: Defining statistical hypotheses

Case study: Opportunity cost

**Class Example 7.2:
Opportunity cost**

	Buy video	Not buy video	Total
Control	56	19	75
Treatment	41	34	75
Total	97	53	150

Class Example 7.3: Opportunity cost, revisited

Building a randomization distribution

The card-shuffling metaphor

Activity

Think of the data as 150 cards:

97 red cards (buy video)

53 white cards (not buy video)

In the actual study, 75 cards were dealt to the 'control' pile and 75 to the 'treatment' pile. Under the null hypothesis, the color of each card (whether to buy the video) has nothing to do with which pile it lands in.

To simulate what the null hypothesis look like, we:

1. *Thoroughly shuffle* all 150 cards together.
2. *Deal 75* into a 'control' pile and 75 into a 'treatment' pile
3. *Count* the number of white (not buy video) in each pile and compute the difference in proportions.

This gives us one simulated difference *under the null hypothesis*, a difference that arose purely from the random shuffle, not from the phrasing.

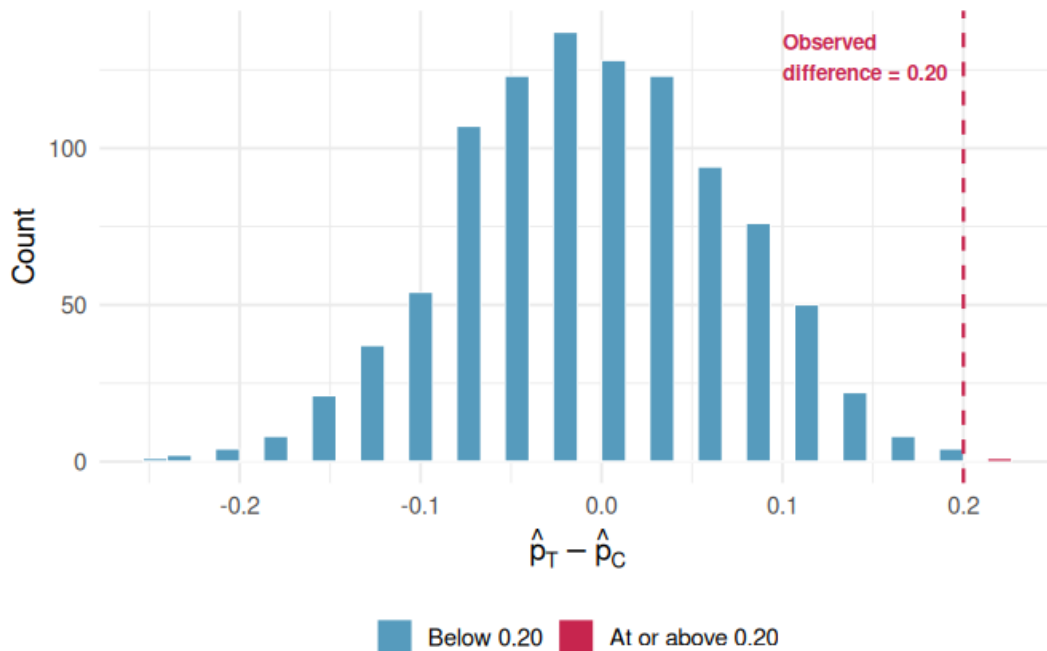


Figure 2: A histogram of 1,000 simulated differences produced under the null hypothesis. Differences at least as large as the observed 0.20 are highlighted in

P-values

Randomization Distributions and p-values

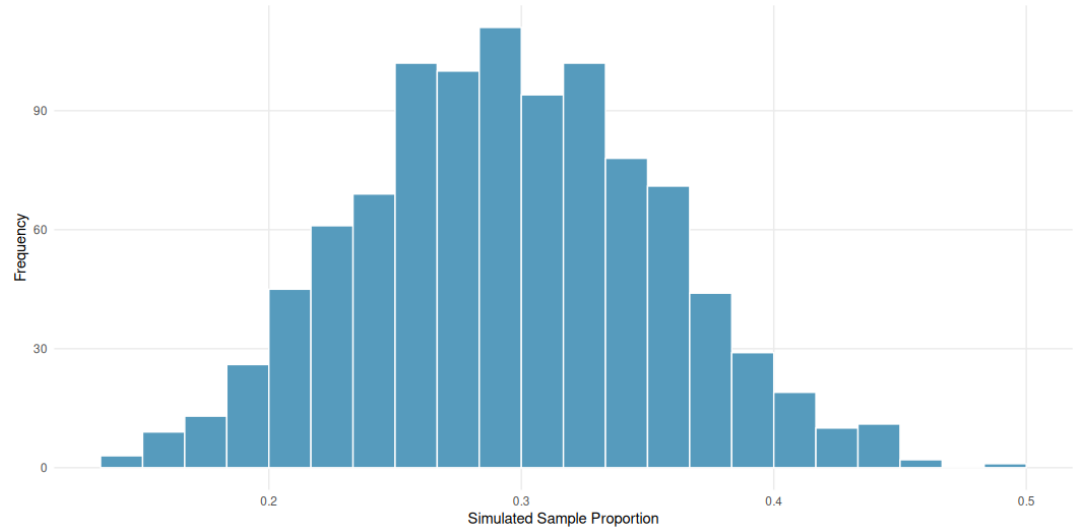


Figure 3: Randomization distribution with 1000 randomizations

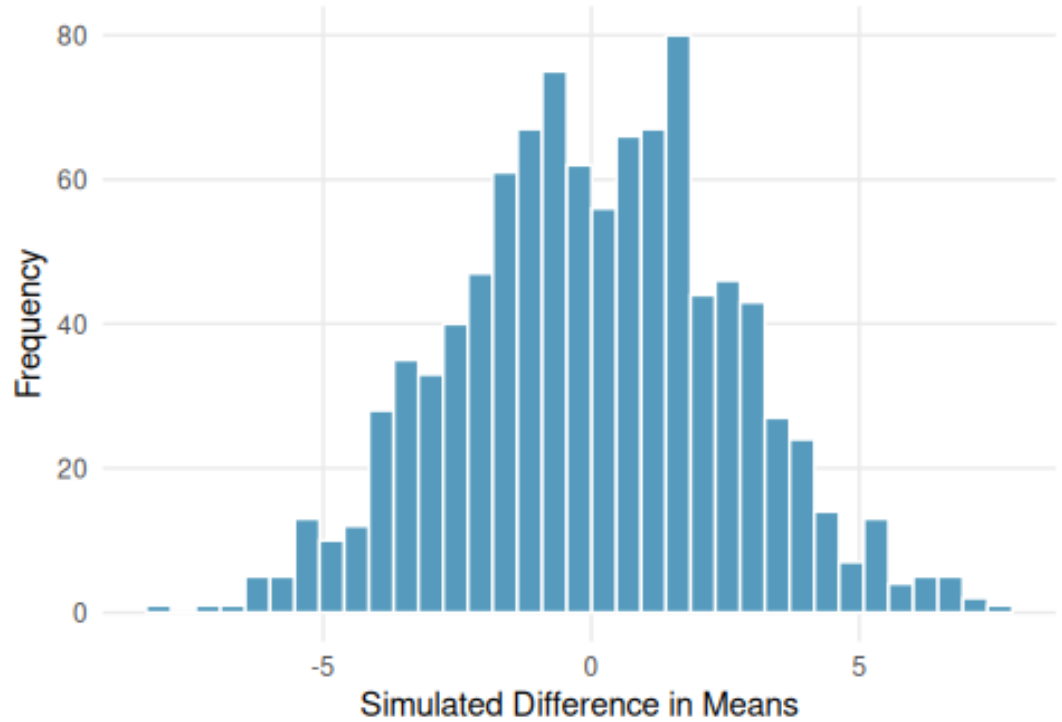


Figure 4: Randomization distribution with 1000 randomizations

Class Example 7.6: Migraine Acupuncture Trial

Class Example 7.7: Calorie Counts

Class Example 7.8: Cereal study revisited

Statistical discernibility

Decision framework

What “statistically discernible” does and does not mean

How small is small enough?

Class Example 7.9: Red wine and weight loss

Class Example 7.10: Red wine and weight loss revisited

Class Example 7.11: Migraine Acupuncture Trial revisited

Class Example 7.12: Calorie count revisited

Class Example 7.13: Cereal study revisited

Type I and Type II errors



Class Example 7.14: Red wine and weight loss revisited

Class Example 4.15: Errors in a drug study

The randomization test procedure: a summary

Summary